



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



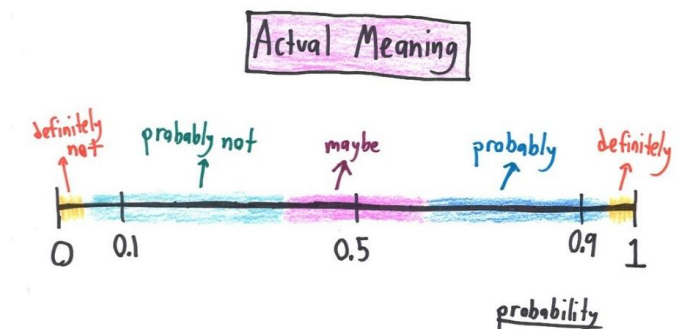


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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Week 14 - Lecture No 26 – November 17, 2025



Announcements!

1. All lectures and grades posted.
2. No class on Wednesday November 19, 2025 - enjoy
3. Quiz No 09 – November 24, 2025 (from Unit 5)

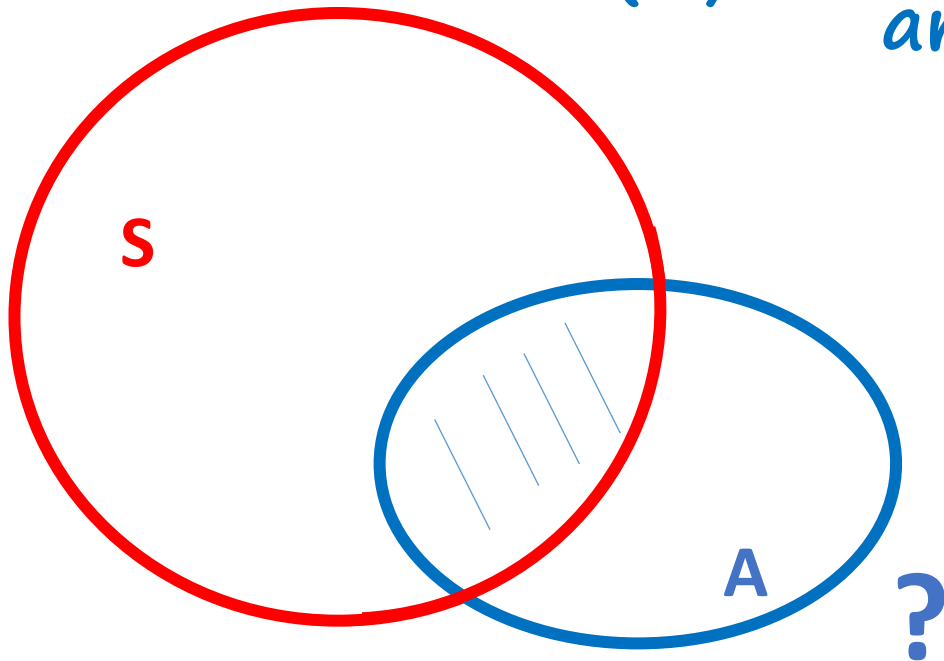
Agenda for today

1. Unit 5 – Continuous Random Variables
 - ➡ ➤ Joint PDFs
 - ➡ ➤ Uniform joint PDF on a set S

Uniform joint PDF on a set S

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\text{area of } S}, & \text{if } (x,y) \in S, \\ 0, & \text{Otherwise.} \end{cases}$$

$$P(A) = \frac{\text{area}(A \cap S)}{\text{area}(S)}$$



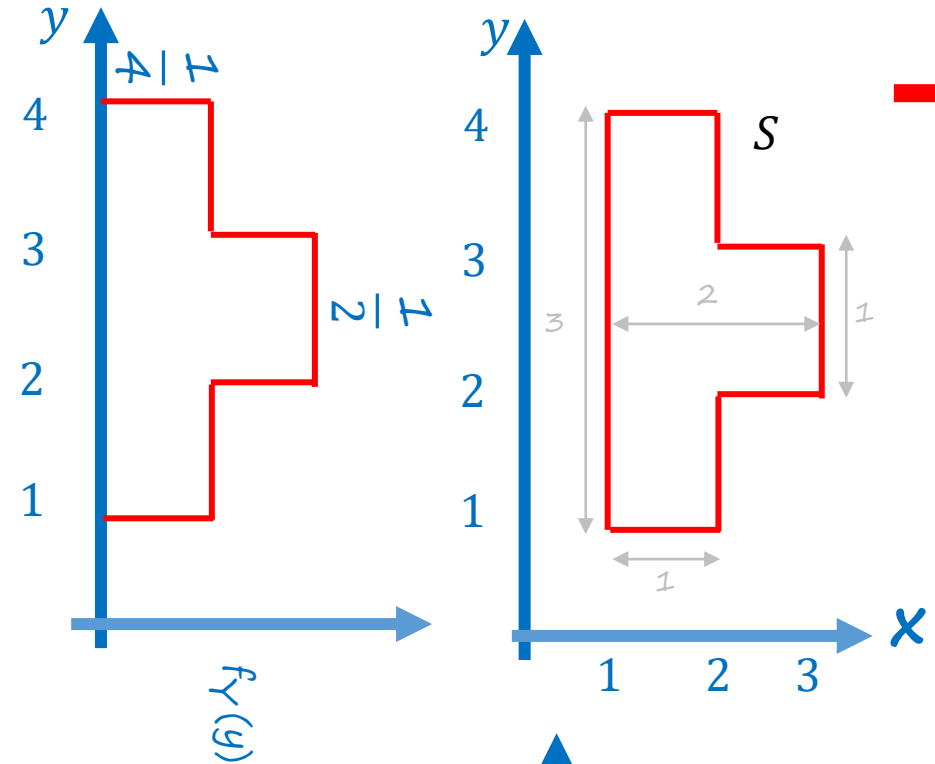
Exercise - Uniform joint PDF on a set S

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\text{area of } S}, & \text{if } (x,y) \in S, \\ 0, & \text{Otherwise.} \end{cases}$$

$$f_X(x) = \int f_{X,Y}(x,y) dy$$

$$f_Y(y) = \int f_{X,Y}(x,y) dx$$

$\rightarrow f_{x,y} = ?$

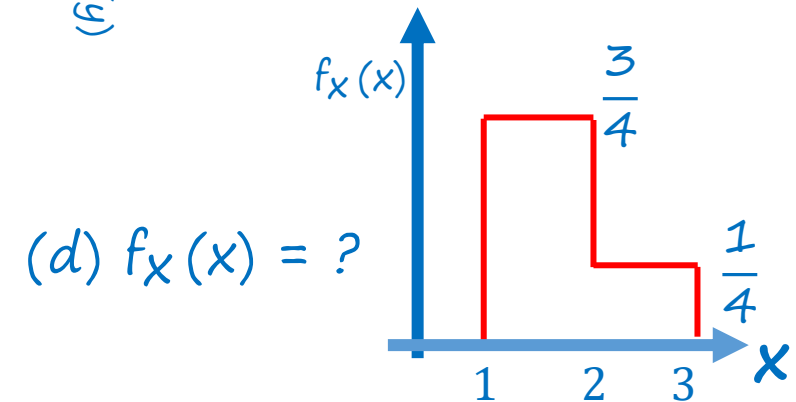


(e) $f_Y(y) = ?$

(a) $f_{X,Y}(4, 4) = ?$

(b) $f_{X,Y}(2, 2) = ?$

(c) $P(1 < X < 2, 1 < Y < 2) = ?$



(d) $f_X(x) = ?$

More than two random variables

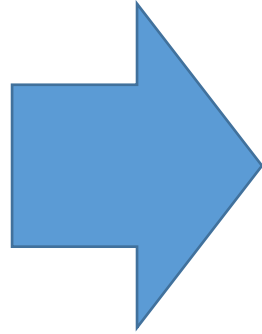
$$\rightarrow p_{X,Y,Z}(x, y, z)$$

$$\rightarrow \sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$$

$$\rightarrow p_X(x) = \sum_z \sum_y p_{X,Y,Z}(x, y, z)$$

$$\rightarrow p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z)$$

$$\rightarrow f_{X,Y,Z}(x, y, z)$$



Sum becomes integrals & pmfs
becomes pdfs

Functions of multiple random variables

$$\rightarrow Z = g(X, Y)$$

Expected value rule:

$$\rightarrow E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y) \quad E[g(X, Y)] = \int \int g(x, y) f_{X,Y}(x, y) dx dy$$

Linearity of expectations

$$\rightarrow E[aX + b] = aE[X] + b$$

$$\rightarrow E[X + Y] = E[X] + E[Y]$$

$$E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$$

The joint CDF

$$\rightarrow F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$\rightarrow f_X(x) = \frac{dF_X(x)}{dx}$$

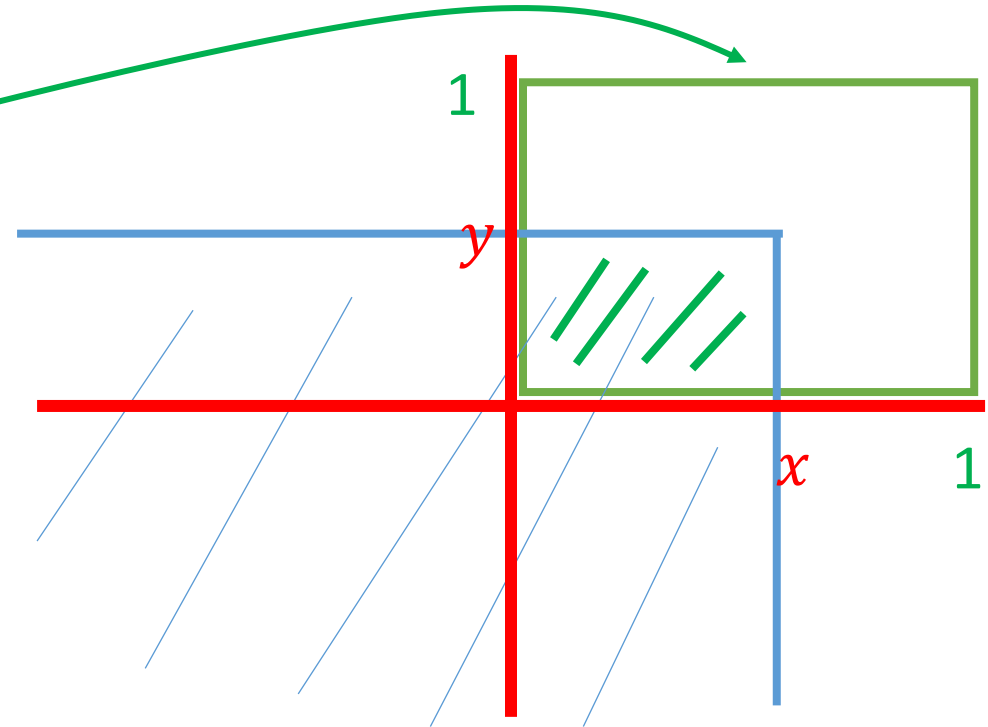
$$\rightarrow F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \left[\int_{-\infty}^x f_{X,Y}(s, t) ds \right] dt$$

$$\rightarrow f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x, y)$$

$$f_{X,Y}(x, y) = 1, \text{ for } 0 < (x, y) < 1$$

$$F_{X,Y}(x, y) = xy$$

$$f_{X,Y}(x, y) = 1$$



Conditional PDFs, given another r.v.

$$p_{X|Y}(x|y) = \mathbf{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} \quad P_y(y) > 0$$

$$p_{X,Y}(x, y) \longrightarrow f_{X,Y}(x, y)$$

$$p_{X|A}(x) \longrightarrow f_{X|A}(x)$$

$$p_{X|Y}(x|y) \longrightarrow f_{X|Y}(x|y)$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{if } f_y(y) > 0$$

$$\mathbf{P}(x \leq X \leq x + \delta \mid A) \approx f_{X|A}(x) \delta \quad \text{where } P(A) > 0$$

if $Y = y$; $P(Y=y) = 0$

So $Y \approx y$

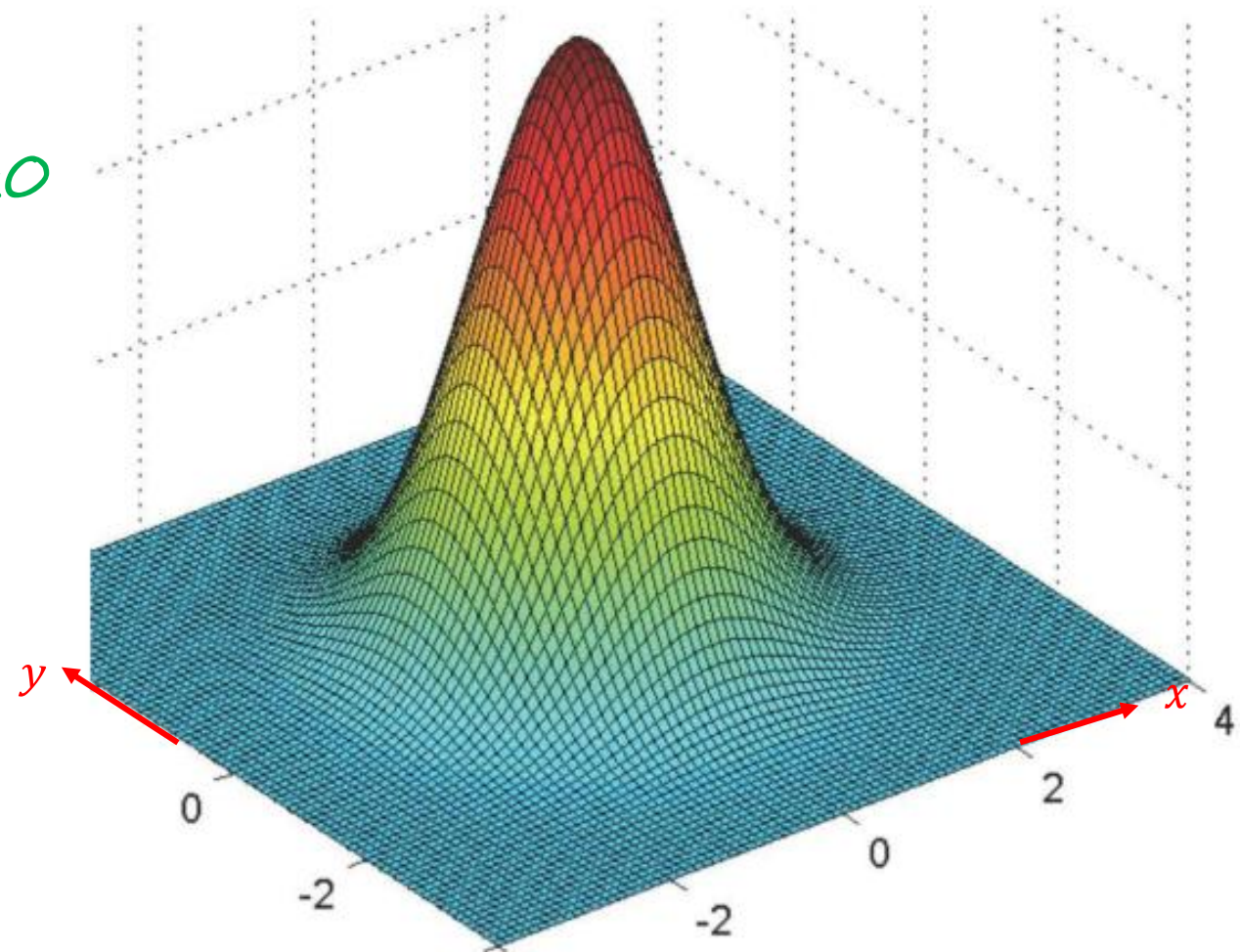
$$\mathbf{P}(x \leq X \leq x + \delta \mid y \leq Y \leq y + \Delta) = \frac{f_{X,Y}(x, y) \delta \Delta}{f_Y(y) \Delta} = f_{X|Y}(x|y) \delta$$

Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y) dx$$

- Some comments:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \bullet \quad f_{X|Y}(x|y) \geq 0$$



Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y) dx$$

- **Some comments:**

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

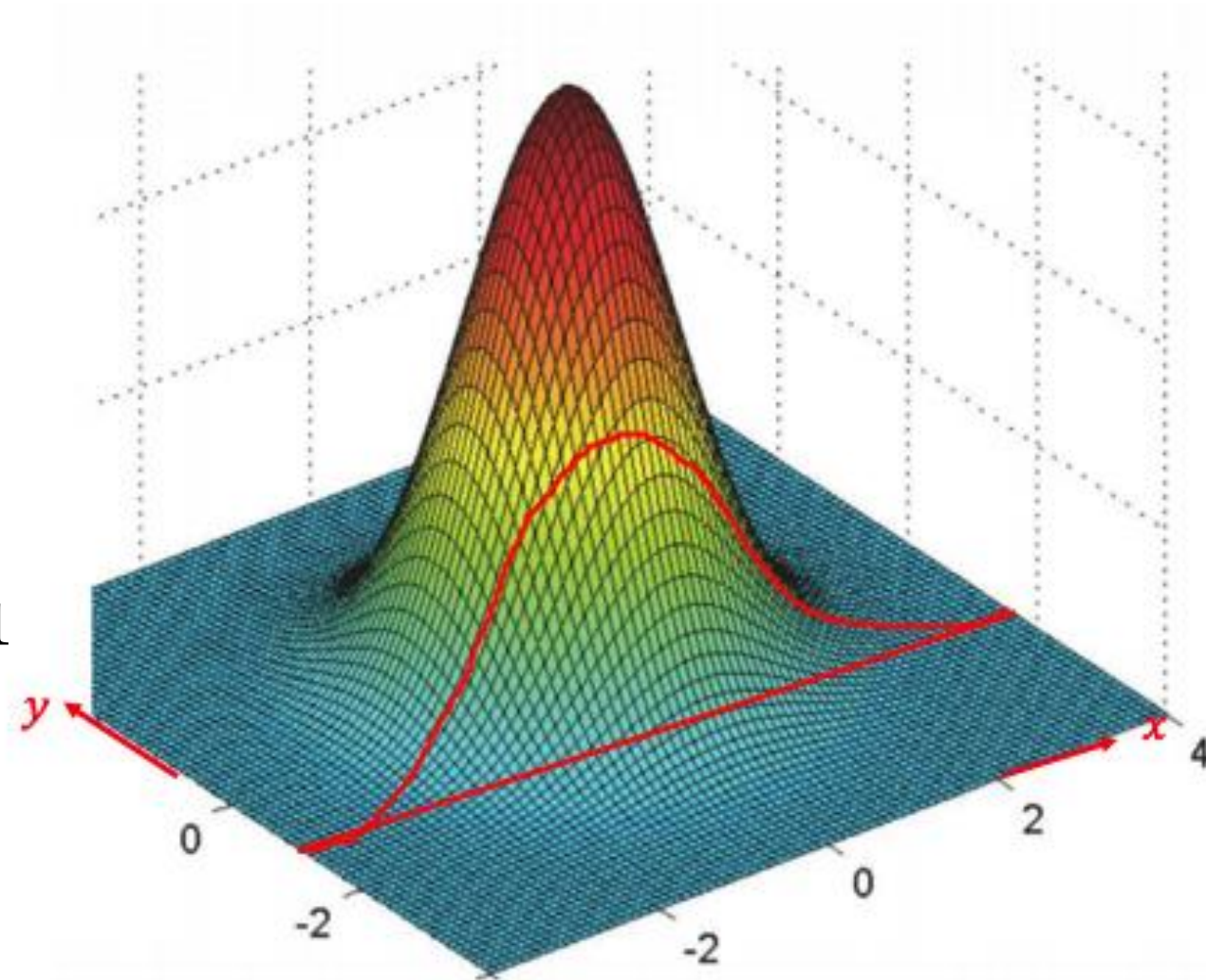
- $f_{X|Y}(x|y) \geq 0$

- Think of the value of Y fixed at y
shape of $f_{X|Y}(\cdot|y)$: slice of the joint

- $\int_{-\infty}^{\infty} f_{X|Y}(x|y) dx = \frac{\int_{-\infty}^{\infty} f_{X,Y}(x,y) dx}{f_Y(y)} = 1$

- Multiplication rule:

$$\begin{aligned} f_{X,Y}(x,y) &= f_{X|Y}(x|y)f_Y(y) \\ &= f_{Y|X}(y|x)f_X(x) \end{aligned}$$



Expected Value Rule

$$\mathbf{E}[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x \mid y)$$



$$E[g(X)|Y=y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y) dx$$

Independence

$$p_{X,Y}(x,y) = p_X(x) p_Y(y) \quad \text{for all } x, y$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \text{for all } x, y$$

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) f_Y(y) \quad \text{always true}$$

$$\rightarrow f_X(x) = f_{X|Y}(x|y) \quad \text{for all } y \text{ with } f_Y(y) > 0 \text{ and all } x$$

$$\rightarrow f_Y(y) = f_{Y|X}(y|x) \quad \text{for all } x \text{ with } f_X(x) > 0 \text{ and all } y$$

Total probability & expectation theorem

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y) \quad \longrightarrow \quad f_X(x) = \int_{-\infty}^{\infty} f_Y(y) f_{X|Y}(x|y) dy$$

$$\mathbf{E}[X | Y = y] = \sum_x x p_{X|Y}(x | y) \quad \longrightarrow \quad E[X | Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

$$\mathbf{E}[X] = \sum_y p_Y(y) \mathbf{E}[X | Y = y] \quad \longrightarrow \quad E[X] = \int_{-\infty}^{\infty} f_Y(y) E[X | Y=y] dy$$

How do you prove the last statement?

Think



Discuss



Prove

Proof

$$\begin{aligned} \mathbf{E}[X] &= \int_{-\infty}^{\infty} f_Y(y) \mathbf{E}[X | Y = y] dy &= \int_{-\infty}^{\infty} f_Y(y) \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx dy \\ & &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_Y(y) f_{X|Y}(x|y) dy dx \\ & &= \int_{-\infty}^{\infty} x \underbrace{\int_{-\infty}^{\infty} f_Y(y) f_{X|Y}(x|y) dy}_{f_X(x)} dx \\ & &= \int_{-\infty}^{\infty} x f_X(x) dx = E[X] \end{aligned}$$

Example

Uniform probabilities on a triangle

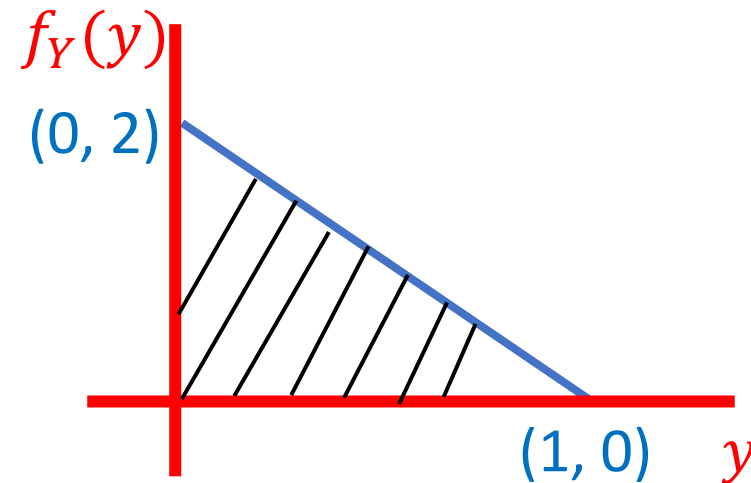
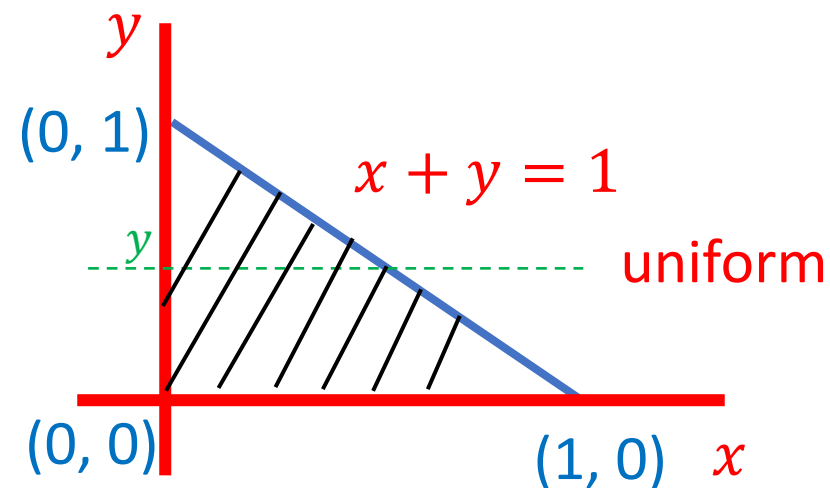
find:

a) $f_{X,Y}(x, y)$

$$f_{X,Y}(x, y) = \begin{cases} 2, & x + y < 1, \\ & x, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

b) $f_Y(y)$

$$\begin{aligned} f_Y(y) &= \int f_{X,Y}(x, y) dx \\ &= \begin{cases} 2(1-y), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$



Example

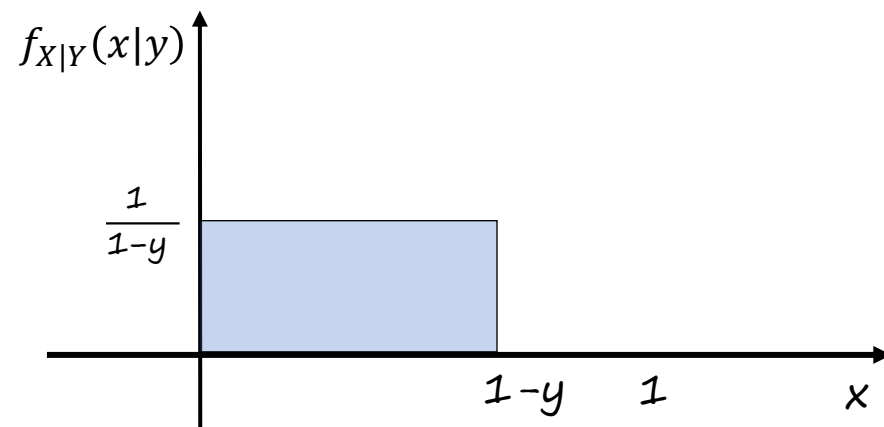
$$c) f_{X|Y}(x|y)$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{2}{2(1-y)}$$

$$= \begin{cases} \frac{1}{(1-y)}, & 0 \leq y \leq 1; 0 \leq x \leq 1-y \\ 0, & \text{otherwise} \end{cases}$$

$$0 \leq y < 1;$$
$$0 \leq x + y \leq 1$$



Example

d) $E[X|Y = y]$

$$E[X | Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

$$E[X | Y=y] = \int_0^{1-y} x \frac{1}{(1-y)} dx$$

$$= \frac{1}{(1-y)} \frac{(1-y)^2}{2}$$

$$= \frac{(1-y)}{2}$$

$$0 \leq y \leq 1$$

Example

e) $E[X]$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} f_Y(y) E[X | Y=y] dy \\ &= \int_{-\infty}^{\infty} f_Y(y) \frac{(1-y)}{2} dy = \frac{1}{2} \int_{-\infty}^{\infty} f_Y(y) dy - \frac{1}{2} \int_{-\infty}^{\infty} y f_Y(y) dy \end{aligned}$$

$$= \frac{1}{2} - \frac{1}{2} E[Y] = \frac{1}{2} - \frac{1}{2} E[X]$$

$$\left(1 + \frac{1}{2}\right) E[X] = \frac{1}{2}$$

$$E[X] = \frac{1}{3}$$

$$E[Y] = \frac{1}{3}$$

Example on Conditional PDFs

- You have two options to play a game of lottery:
 - ❖ Option 1: Select $X \sim U(0, 100)$ then select $Y \sim U(0, X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$
 - ❖ Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$

Which option should you select?



Example on Conditional PDFs

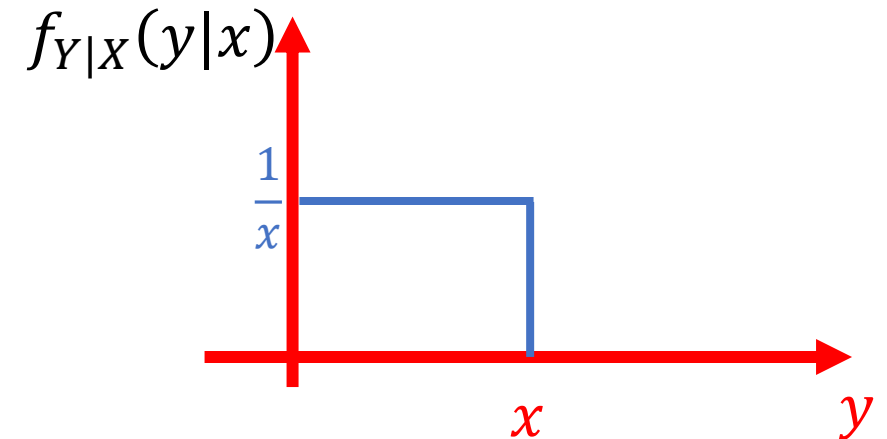
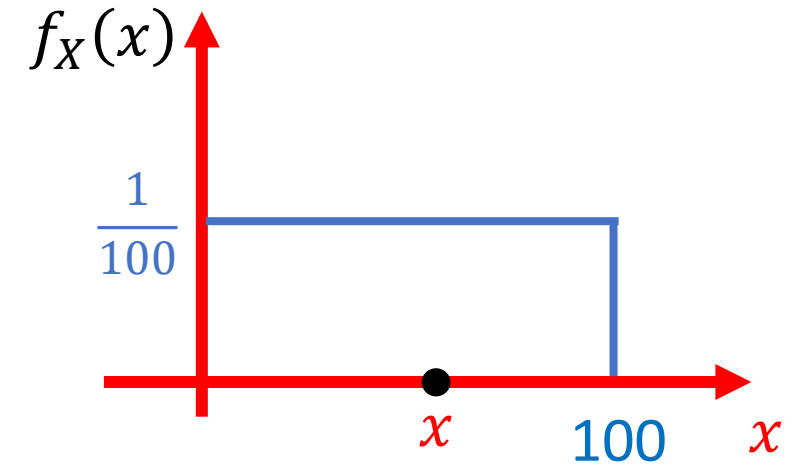
Lets analyze option 1:

Select x , where $X \sim U(0, 100)$ then select y such that $Y \sim U(0, X)$.
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Option 1: Select x , where $X \sim U(0, 100)$ then select y such that $Y \sim U(0, X)$.
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Are X & Y independent? If $x = 0.5$, this implies $y \leq 0.5$

$$f_{X,Y}(x, y) = f_{Y|X}(y|x)f_X(x)$$

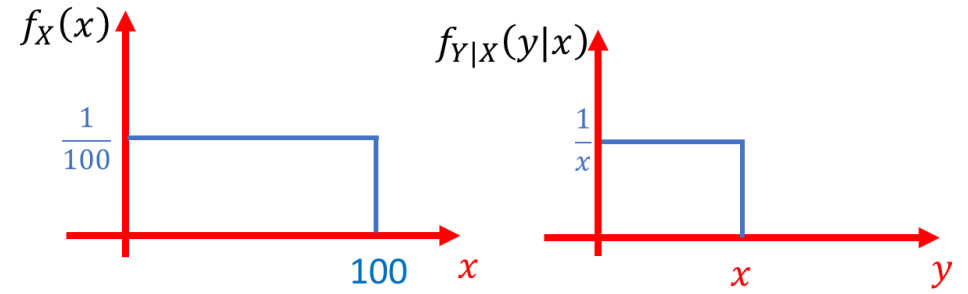


Option 1: Select $X \sim U(0, 100)$ then select $Y \sim U(0, X)$. You win Y rupees.

Find $f_Y(y)$ and $E[Y]$

$$f_{X,Y}(x, y) = f_{Y|X}(y|x)f_X(x)$$

$$\rightarrow = \frac{1}{100} * \frac{1}{x} \quad 0 \leq y \leq x \leq 100$$

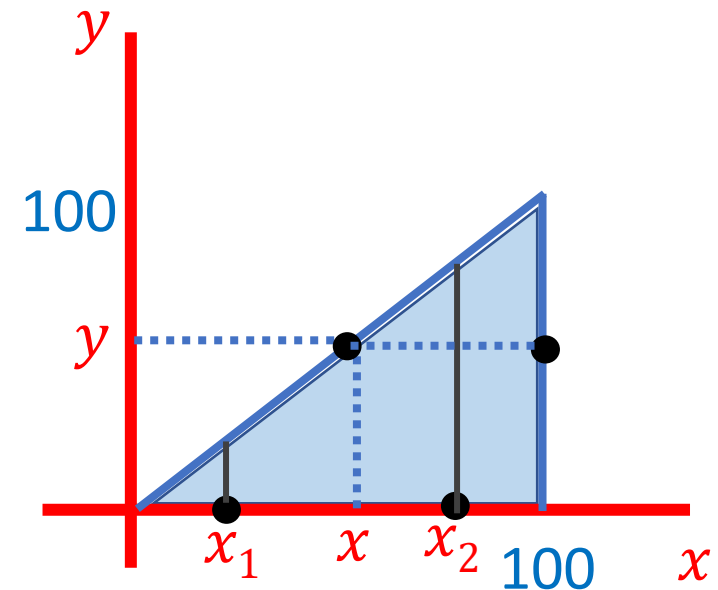


$$\rightarrow f_Y(y) = \int_x f_{X,Y}(x, y) dx = \int \frac{1}{100} * \frac{1}{x} dx$$

$$\rightarrow = \frac{\ln x}{100} \Big|_y^{100} = \frac{1}{100} \ln(100/y) ?$$

$$\rightarrow E[Y] = \int_0^{100} y \frac{1}{100} \ln(100/y) dy$$

$$\rightarrow = \int_0^{100} E[Y|X=x] f_X(x) dx = \int_0^{100} \frac{1}{100} * \frac{x}{2} dx = 1/2 E[X] = 25$$



Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees.
Find $f_Y(y)$ and $E[Y]$

$$f_{X,Y}(x,y) = f_{Y|X}(y|x)f_X(x)$$

$$\rightarrow = \frac{1}{50} * \frac{1}{2x} = \frac{1}{100x} \quad 0 \leq x \leq 50; 0 \leq y \leq 2x$$

$$\rightarrow f_Y(y) = \int_x f_{X,Y}(x,y) dx = \int_{y/2}^{50} f_{X,Y}(x,y) dx \quad 0 \leq y \leq 100$$

$$\rightarrow = \frac{\ln x}{100} \Big|_{y/2}^{50} = \frac{1}{100} \ln(100/y)$$

$$\rightarrow E[Y] = \int_0^{50} y \frac{1}{100} \ln(100/y) dy$$

$$\rightarrow = \int_0^{50} E[Y|X=x] f_X(x) dx = \int_0^{50} \frac{1}{50} x dx = 1/2 E[X] = 25$$

